

Data Scientist

Routing Optimization · Mobility Intelligence · Trajectory Analytics · AI-assisted Map Quality

 Tasco Building, Hanoi – Pham Hung

 TASCO Mapping

 Individual Contributor

 Reports to Head of Mapping / DS Lead

ABOUT THE COMPANY

About Tasco

VISION

A technology-enabled investment ecosystem, integrating Smart Mobility, Insurance, and Investments in essential services and innovation.

STRATEGIC PARTNERS

Geely Auto

Mitsui & Co.

IFC – World Bank Group

Corporate Information

Stock Ticker	HUT (HNX)
Established	1971
Subsidiaries & Affiliates	88
Employees	~10,000
Consolidated Sales	\$1.4 billion
Total Assets	\$1.98 billion
Website	tasco.com.vn

BUSINESS ECOSYSTEM

MOBILITY

Tasco Auto
VETC
Carpla
VETC Go

INSURANCE

Tasco Insurance
Tasco Finance

INVESTMENT

DNP Water · Tasco Mall
Ana Mandara Villas
Six Senses
Inochi

WHY TASCO

Our Mobility Ecosystem in Numbers

TASCO AUTO

40k+ **180** **Top 5** **16** **14.8%**
New Cars Sold Showrooms Tax-paying Retailers Car Brands Market Share

VETC — ETC & MOBILITY

4.2m **~300** **2m** **650k** **75%**
Car Owners Parking Lots Transactions / Day Car Repairs ETC Market Share

0 Job Overview

POSITION	Data Scientist
FOCUS	Routing Optimization, ETA, Trajectory Mining, Driver Preference Modeling, Map & Mobility Data Science
DEPARTMENT	TASCO.COM.VN Mapping
LOCATION	Tasco Building Hanoi, Pham Hung
LEVEL	Individual Contributor
REPORTS TO	Head of Mapping / Mapping Data Science Lead

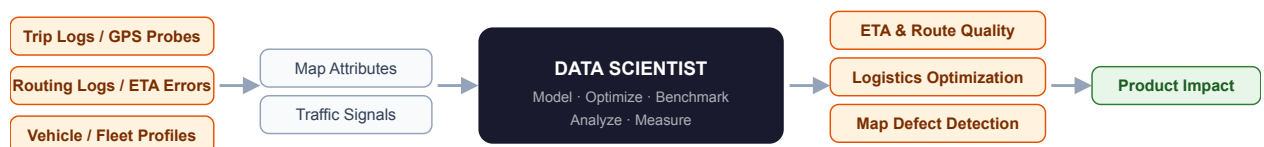
1 Job Purpose

The Data Scientist supports the development of data-driven routing, navigation, logistics, and mobility intelligence systems for Tasco Maps, VETC, MyTasco, fleet services, tolling, parking, insurance, logistics, and future automotive products.

This role converts GPS probes, trip logs, routing results, ETA errors, traffic signals, vehicle profiles, map attributes, and user behavior into measurable models and optimization insights.

The position is **not limited to dashboarding**. It is designed for a candidate who can gradually grow into building Vietnam-localized routing intelligence: time-dependent routing, route quality evaluation, driver preference modeling, logistics assignment, and AI-assisted detection of routing / map defects.

HOW THE ROLE WORKS — MOBILITY DATA TO INTELLIGENCE



2 Key Responsibilities

A. Routing & ETA Data Science

- Analyze routing logs, trip trajectories, GPS probes, route alternatives, actual-vs-planned paths, and ETA prediction errors.
- Build features for ETA models using road class, time of day, congestion level, toll road usage, intersection complexity, turn penalty, speed profile, event signals, and vehicle type.

- Evaluate route quality using actual travel time, distance, delay, detour ratio, rerouting frequency, user cancellation, toll cost, violation risk, and driver adherence.
- Build dashboards and automated reports for ETA error, routing failure, road-segment speed anomaly, and map-data regression.
- Work with Engineering to improve dynamic costing inputs for routing engines such as Valhalla, OSRM, or GraphHopper.

B. Vehicle Routing & Logistics Optimization

- Support modeling of logistics and fleet routing problems: vehicle routing, pickup-and-delivery, capacity constraints, time windows, heterogeneous vehicle types, driver working hours, depot constraints, and service time.
- Prototype optimization models using Python and open-source solvers for small to medium-sized planning problems.
- Compare exact optimization, heuristic, local search, and rule-based approaches for real-world dispatching and routing use cases.
- Build benchmark datasets for Vietnamese use cases: toll-road logistics, car-service pickup/dropoff, roadside assistance, EV charging support, vehicle inspection routing, and last-mile delivery.

C. Time-dependent Travel Time & Traffic Intelligence

- Build time-dependent travel-time profiles by road segment, road class, city, day of week, and time window.
- Detect recurring congestion patterns, abnormal delay, road closure signals, and travel-time instability from GPS probe data.
- Support construction of piecewise travel-time / speed functions for dynamic routing and simulation.
- Work with MapOps and Engineering to identify roads where missing restrictions, wrong road class, wrong speed, or bad topology cause routing degradation.

D. Personalized Route Planning & Driver Preference Modeling

- Analyze actual route choices to infer driver, vehicle, and fleet preferences: fastest route, toll-avoidance, highway preference, safe route, familiar route, low-congestion route, or compliance-first route.
- Build preference features and clustering models for user groups: private car users, taxi drivers, logistics fleets, bus / coach drivers, EV drivers, and toll-road users.
- Support alternative-route ranking: not only fastest route, but also safest, most reliable, lowest toll, lowest violation risk, or most familiar.
- Segment real-world trajectories into meaningful sub-trips and infer intermediate destinations or route-choice intent.

E. Trajectory Mining & Map Quality Detection

- Mine GPS trajectories to detect map defects: missing roads, wrong turn restrictions, wrong one-way tags, incorrect toll access, bad entrances, wrong road-side access, and routing through prohibited roads.
- Compare planned route vs actual route to identify systematic driver deviation.

- Build anomaly detection for map-matching errors, GPS drift, impossible turns, route snapping failure, and inconsistent speed profiles.
- Generate prioritized MapOps review queues from data signals: high-traffic, high-impact, high-confidence, and high-risk map issues.

F. Experimentation, Benchmarking & Model Evaluation

- Design experiments with clear baseline, treatment, success metric, guardrail metric, and rollout criterion.
- Build offline benchmarks for ETA, route ranking, travel-time prediction, route adherence, trajectory segmentation, and map-defect detection.
- Evaluate model performance using MAE, MAPE, RMSE, precision, recall, F1, calibration, lift, cost reduction, SLA compliance, and operational review cost.
- Conduct error analysis and identify whether issues come from data sparsity, noisy GPS, map error, routing-engine behavior, model bias, or product UX.

G. Documentation, SOPs & Team Enablement

- Build documentation for metric definitions, feature specifications, experiment reports, model cards, RCA notes, and benchmark datasets.
- Translate analytical findings into concrete MapOps, Engineering, Product, or Business actions.
- Support junior team members in SQL, Python analysis, geospatial data processing, and experiment design.

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Requirements

MUST-HAVE

- ✓ Bachelor's degree in Data Science, CS, Applied Mathematics, Operations Research, Statistics, Transportation Engineering, EE, GIS, or related field.
- ✓ 0–2 years in data science, mobility analytics, routing, logistics, map data, optimization, or machine learning.
- ✓ Strong Python skills for data analysis, modeling, and experiment automation.
- ✓ Solid SQL skills; comfortable with large trip tables, GPS logs, routing logs, and geospatial joins.
- ✓ Basic knowledge of ML: regression, classification, clustering, model validation, train/test split, leakage prevention, and error analysis.
- ✓ Basic understanding of graph concepts: nodes, edges, shortest path, route cost,

NICE-TO-HAVE

- ✓ Knowledge of operations research: LP, mixed-integer programming, VRP, TSP, pickup-and-delivery, time windows, local search, and heuristics.
- ✓ Experience with OR-Tools, PuLP, Pyomo, Gurobi, CPLEX, SCIP, NetworkX, igraph, or similar tools.
- ✓ Experience with geospatial tools: PostgreSQL/PostGIS, QGIS, GeoPandas, Shapely, H3, DuckDB, OSMnx, Valhalla, OSRM, GraphHopper.
- ✓ GPS trajectory processing, map matching, route similarity, Fréchet distance, dynamic time warping, or trajectory clustering.
- ✓ Familiarity with streaming / big data tools: Kafka, Flink, Spark, Airflow, MinIO, or data lake pipelines.

road network, trajectory, and map matching.

- ✓ Ability to work with messy mobility data: noisy GPS, sparse probes, inconsistent timestamps, missing labels, outliers, and biased samples.
- ✓ Strong English reading ability for technical papers and routing / optimization documentation.

- ✓ GitHub portfolio, Kaggle work, university thesis, optimization project, logistics analytics project, or map/routing notebook.

SOFT SKILLS & MINDSET

- ✓ Mathematical and data-driven mindset: able to convert an operational problem into variables, constraints, objectives, and metrics.
- ✓ Skeptical about data quality; does not trust GPS logs, map data, or model output without validation.
- ✓ Comfortable with ambiguity and imperfect real-world systems.
- ✓ Practical orientation: cares about product impact, computation cost, latency, and human review effort.
- ✓ Interested in building Vietnam-specific mobility intelligence rather than only applying generic models.
- ✓ Good balance between research curiosity and production discipline.

4 Success Metrics — 6 to 12 Months

- ♦ Build a baseline ETA / travel-time error dashboard for priority cities and road classes.

- ♦ Deliver one offline benchmark for route quality evaluation: planned route vs actual trajectory, ETA error, route adherence, reroute rate, and detour ratio.

- ♦ Build one prototype for detecting map/routing defects from GPS trajectory deviations.

- ♦ Build one route-choice or driver-preference clustering analysis from real trip data.

- ♦ Support one logistics / fleet optimization prototype with capacity, time window, vehicle type, and service-time constraints.

- ♦ Reduce manual MapOps review effort by ranking high-confidence routing/map issues.

- ♦ Define success and guardrail metrics for at least one routing experiment.

- ♦ Produce clear internal documentation: metric spec, feature spec, model report, error taxonomy, and next-step recommendation.

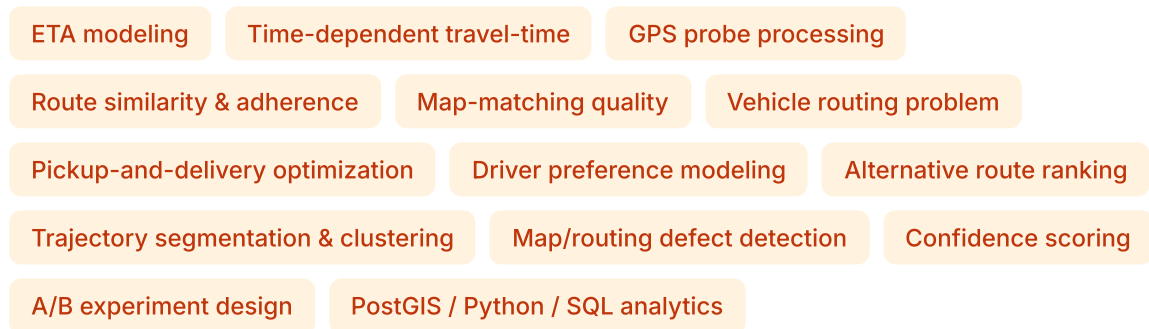
5 Product Scope

Product Line

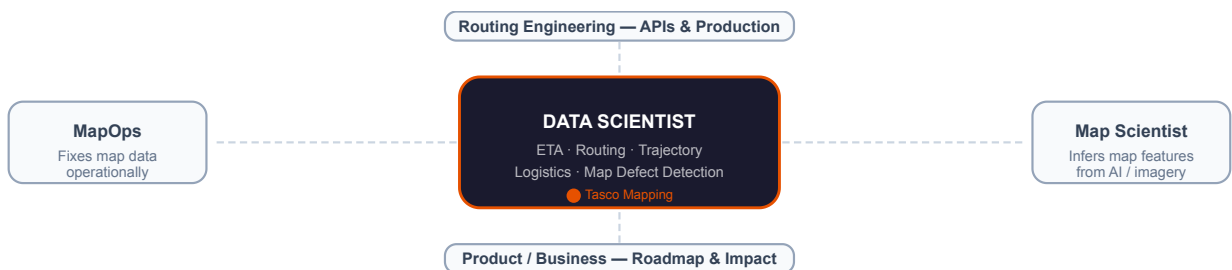
Responsibility

V-Navigation-Maps	ETA error, route quality, route adherence, dynamic costing signals
V-Base-Maps	Data quality metrics, road attribute confidence, map-defect detection
Mobility Data Platform	GPS probe analytics, trajectory mining, traffic profiles
Fleet / Logistics	Vehicle routing, dispatch optimization, time-window feasibility
VETC / Toll Mobility	Toll-road route choice, toll access, toll-station travel-time patterns
Safety / Compliance Navigation	Violation-risk route scoring, restricted-road deviation analysis
Parking / EV / Service Network	Destination routing, service-area routing, vehicle-service dispatch
MapOps QA Platform	AI-assisted prioritization, anomaly detection, review queue scoring

6 Technical Working Areas



7 How This Role Differs



Role	Main Ownership
Map Operations Specialist	Validates and corrects map data manually / operationally
Map Scientist	Builds AI/geospatial methods to infer map features from imagery, OSM, GPS, and field data
AI / Routing Engineer	Builds model infrastructure, routing engine, APIs, graph building, and production services
Data Scientist	Builds ETA, routing, trajectory, logistics models and benchmarks

*MapOps fixes map data. Routing Engineering serves routes. Map Scientist infers map features. **Data Scientist** measures, models, and optimizes the behavior of trips, routes, drivers, fleets, and ETA at scale.*

How to Apply

Send your email with subject **[Tasco Mapping Application - Data Scientist]** with your CV and a portfolio or description of relevant projects in data science, routing, ETA, logistics optimization, GPS trajectory analysis, map matching, transportation analytics, or machine learning.

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[Tasco Mapping Application – Data Scientist]

GitHub repositories, notebooks, route optimization demos, SQL analysis samples, mobility dashboards, academic thesis work, or technical reports are encouraged.